

STUDENT ATTENDANCE MANAGEMENT SYSTEM



Group Members:

MUHAMMAD UZAIR IQBAL (2501469)

MUHAMMAD ABDULLAH KHAN (2501475)

MUHAMMAD SOBAN AKHTAR (2501499)

BE MECHATRONICS (Session 2025-2026)

Project Supervisor

Mr. Umer Farooq Qureshi

Professor

DEPARTMENT OF MECHATRONICS ENGINEERING

FACULTY OF ENGINEERING

AIR UNIVERSITY, ISLAMABAD

Team Member Name	Role And Color	Word count that each has written in assigned color code in report
Muhammad Soban Akhtar	Team Lead	744 words
Muhammad Abdullah Khan	Project Manager	730 words
Muhammad Uzair Iqbal	Technical Integrator and Tester	750 words

Chapter 1- Preliminaries

Proposal:

Student attendance management system is an application software developed to automate the process of recording and managing attendance of students. Attendance system is based on manual record keeping for overall attendance process. It results in errors in recording data and poor management of data. The suggested system provides computerized attendance management which stores the records systematically and retrieves the data whenever alteration is needed. It has capability to add new student records, search records, modify attendance details, delete records and view full attendance details. By switching from a manual attendance register to a digital attendance register you can now take accurate, paperless attendance, eliminating the time, effort and the unnecessary use of resources in managing the attendance system.

Initial Feasibility:

Technical Feasibility:

This project can be implemented using standard C++ programming language and file handling techniques. The software tools needed are free and work on most computer systems.

Economic Feasibility:

This project contains no additional investment on hardware because it uses computer systems and free development environments

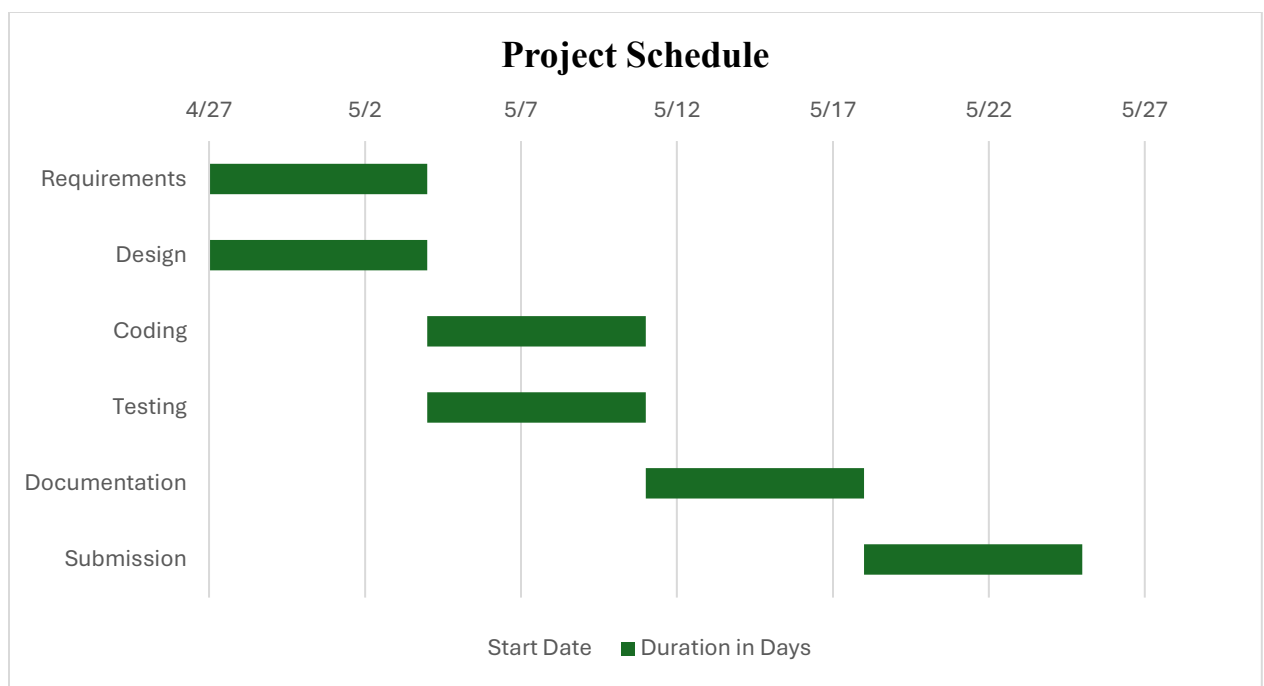
Team Roles And Details:

Member	Role
Muhammad Soban Akhtar	Team Lead
Muhammad Abdullah Khan	Project Manager
Muhammad Uzair Iqbal	Technical Integrator & Tester

Work Breakdown Structure:

- Requirement Analysis
- System Design
- Database Design
- Coding and Implementation
- Testing
- Documentation
- Final Submission

Gantt Chart:



Estimated Budget:

Item	Cost (PKR)
ESP32	1300

Chapter 2 – Project Conception

Introduction:

Attendance management is a key feature of educational institutions to monitor student participation and academic involvement. Attendance has traditionally been maintained manually, either via registers or paper-based systems. Such methods are time-consuming, prone to human error and difficult to maintain over long periods. To overcome these limitations computerized attendance management system is developed. Student Attendance Management System is a console-based application developed in C++ to automate the process of recording attendance of students. The system uses CSV file handling for permanent storage of information and has efficient ways of managing records. Through this project user can add new student record, search existing record, update attendance information, delete record and view complete attendance details.

Project Development Process:

The development of the Student Attendance Management System followed the following process

Requirement Analysis:

During the process the project requirements were identified. The system needed to support registration of student, attendance management, searching of record, updating of record, deletion of record and showing all the records.

System Design:

A structure designed in which each operation is applied separately this improves stability maintainability and readability

Coding and Implementation:

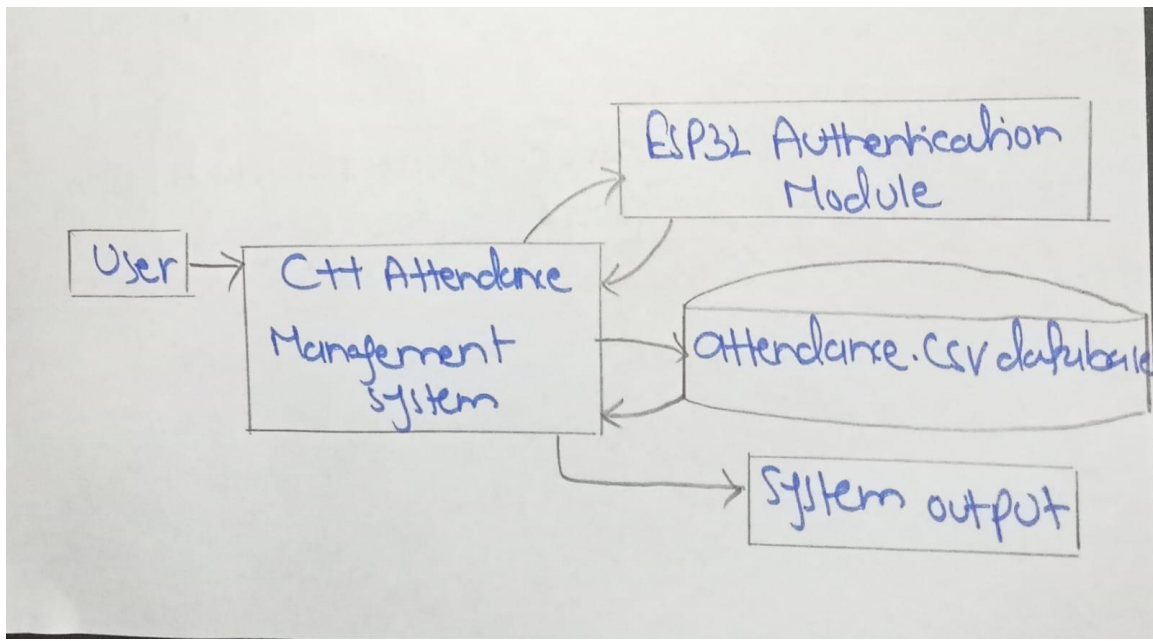
The software was implemented in C++ using:

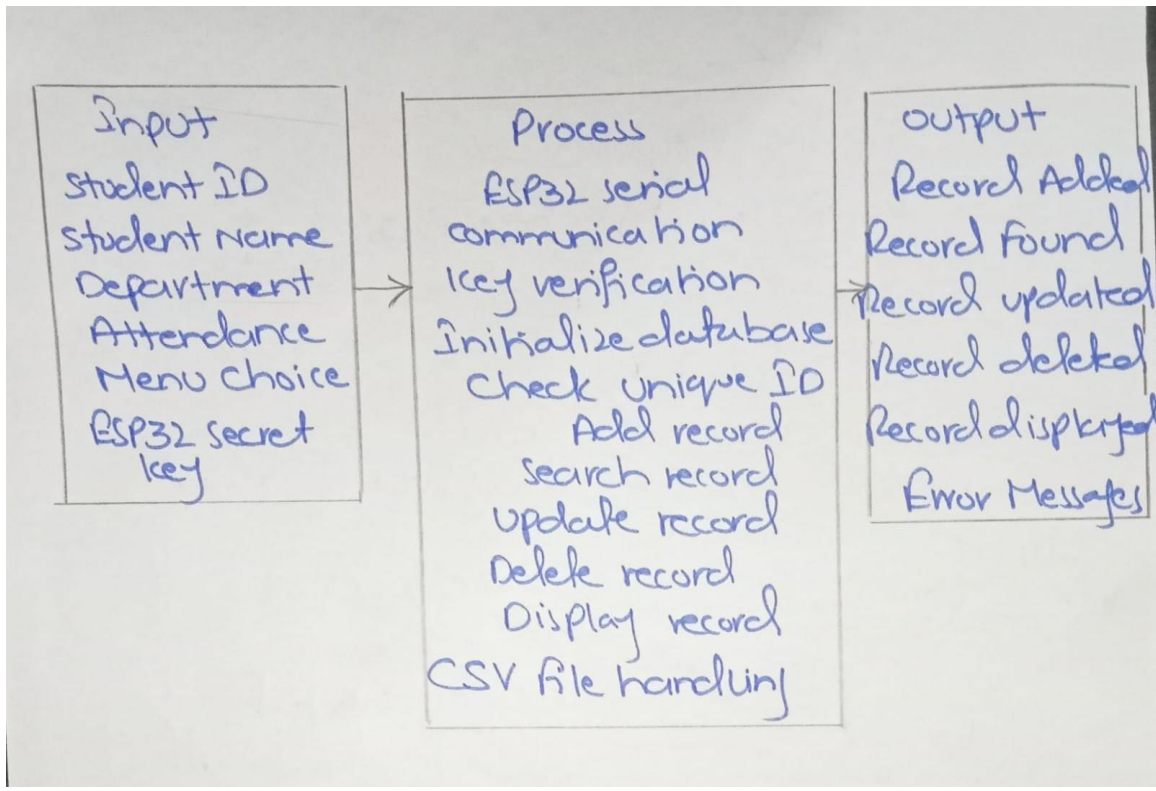
- Functions
- Conditional statements
- Loops
- String manipulation
- File handling operations

Testing:

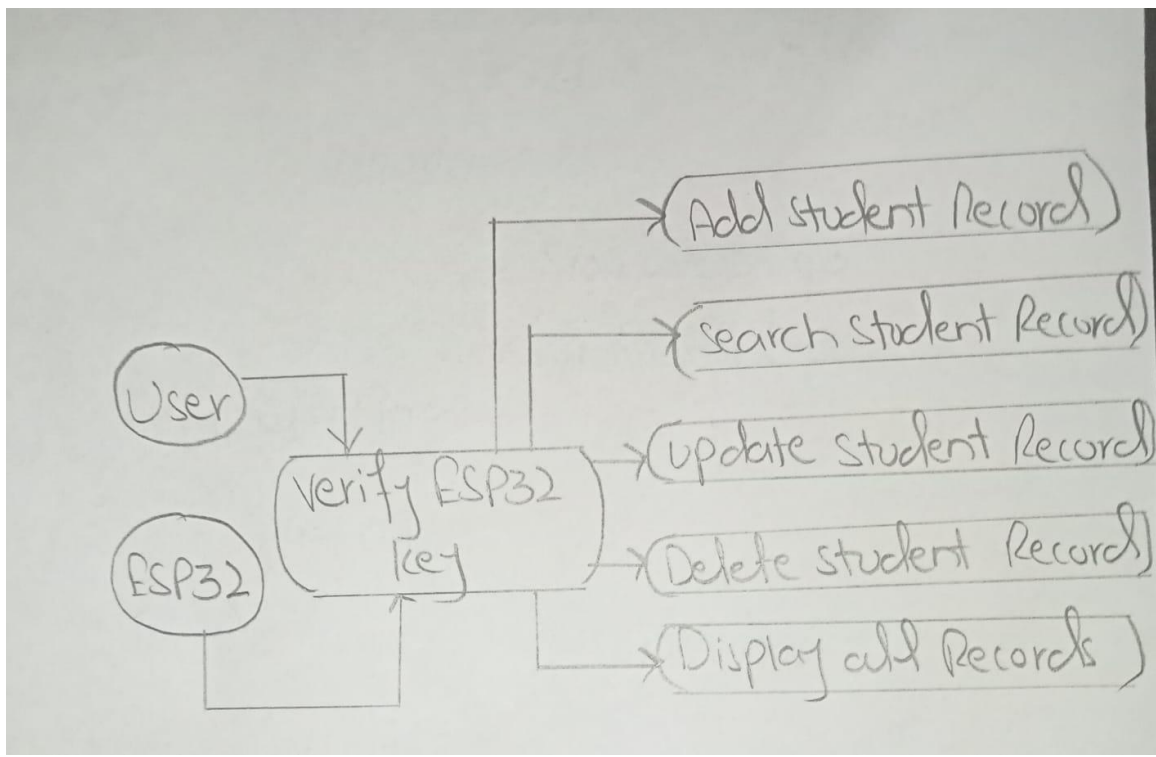
The system was tested independently to make sure the code is correct before finalizing the code

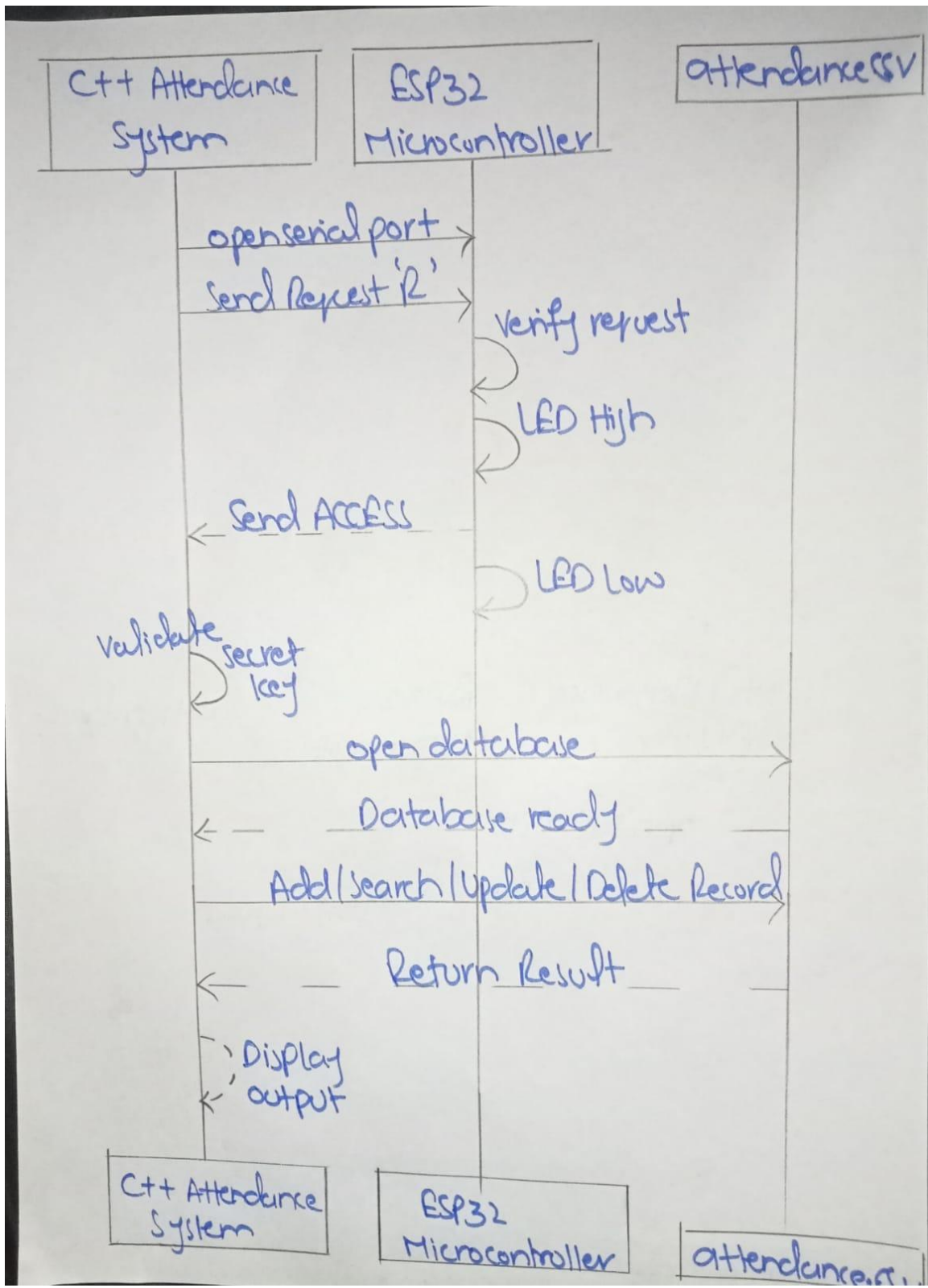
Block Diagram of the System:



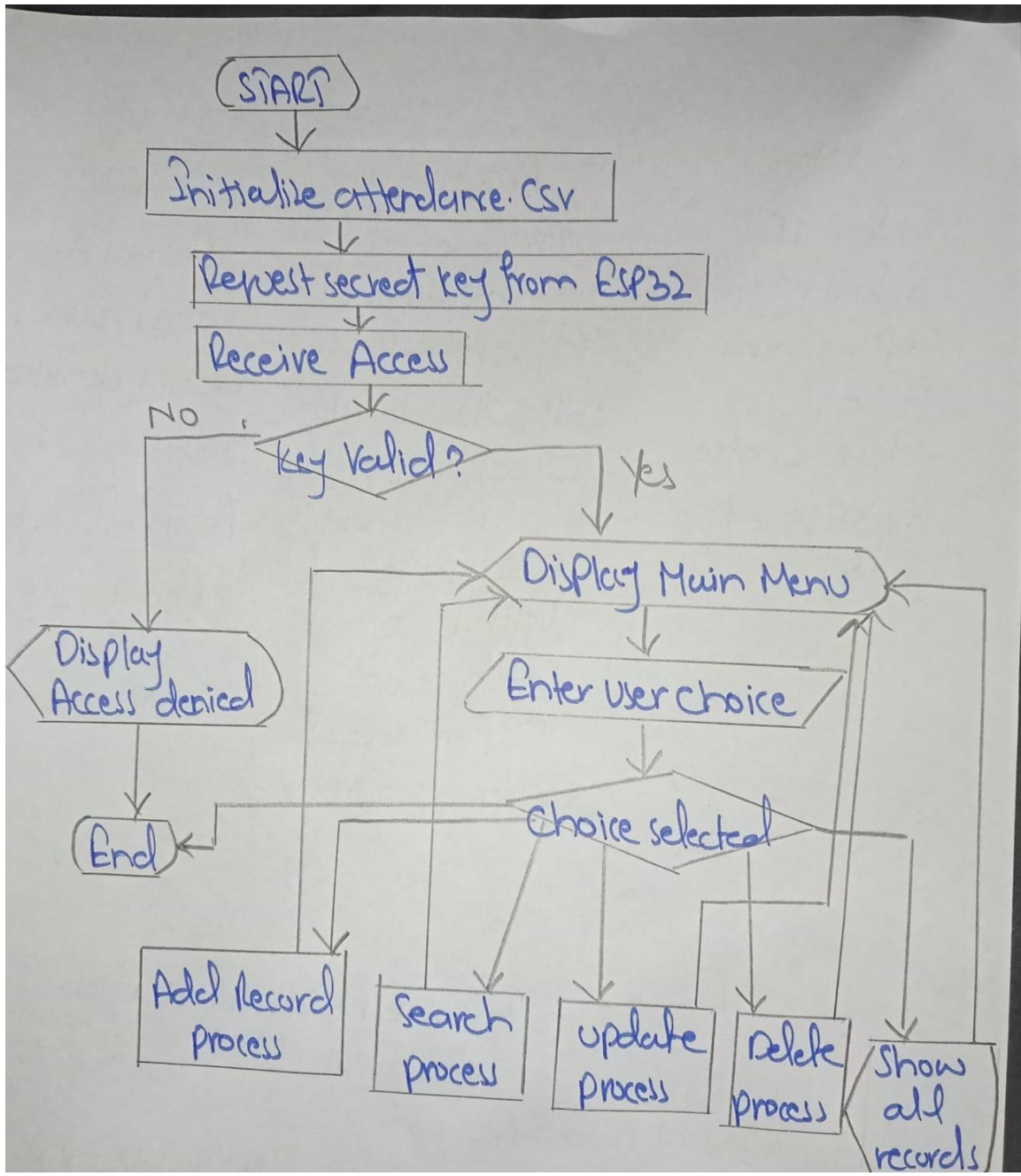


UML Diagram of the System:

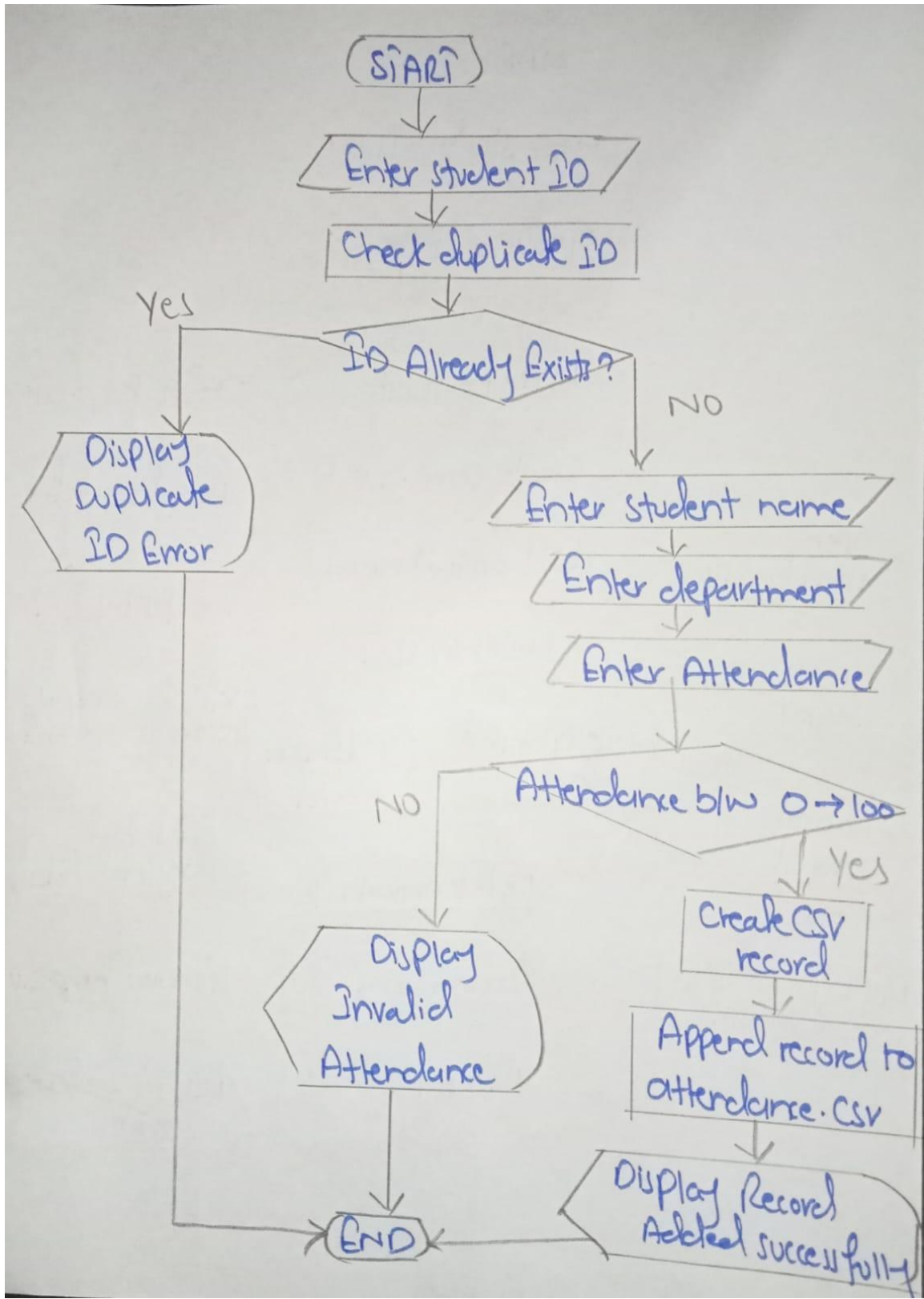




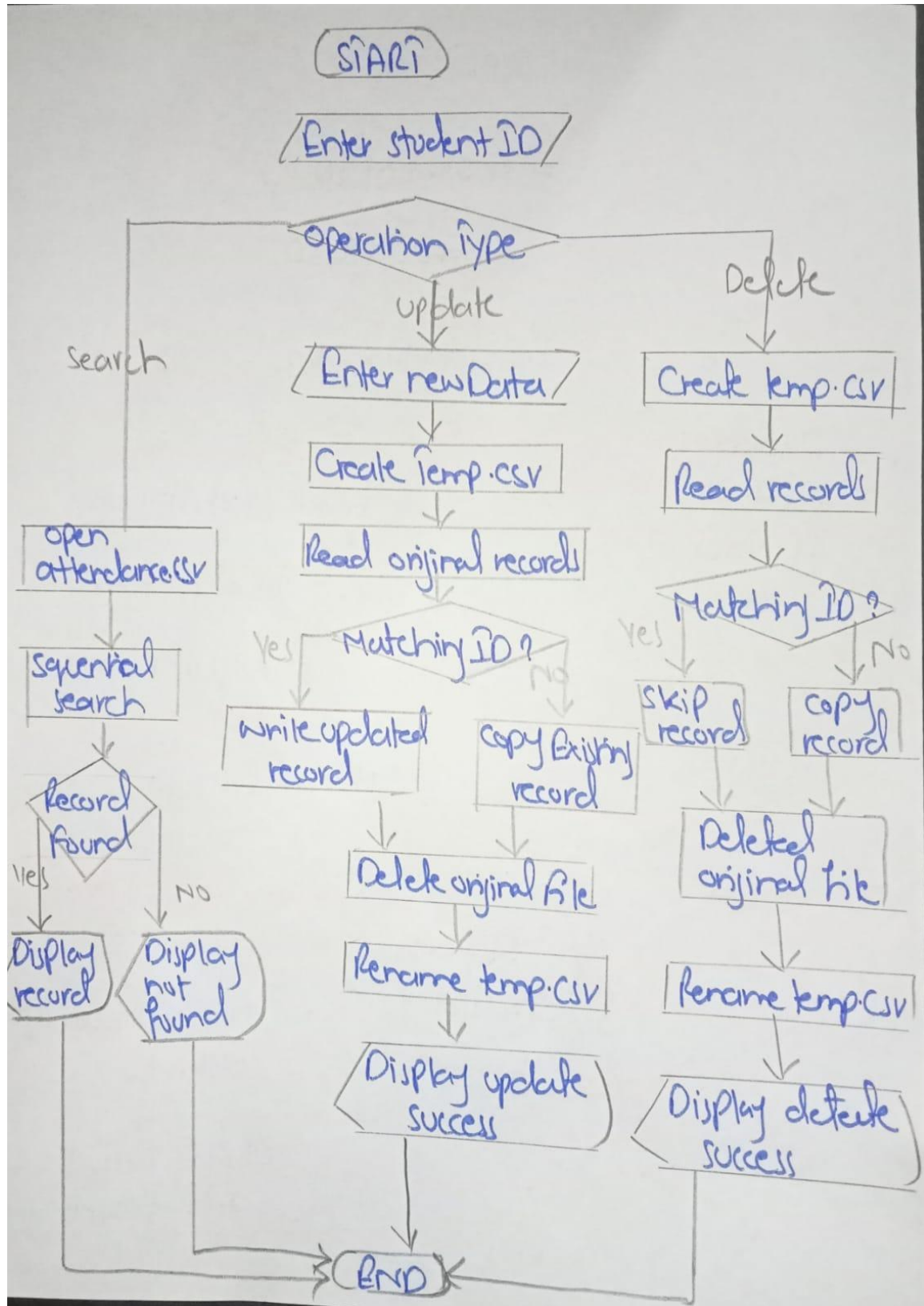
Flowchart for Main System:



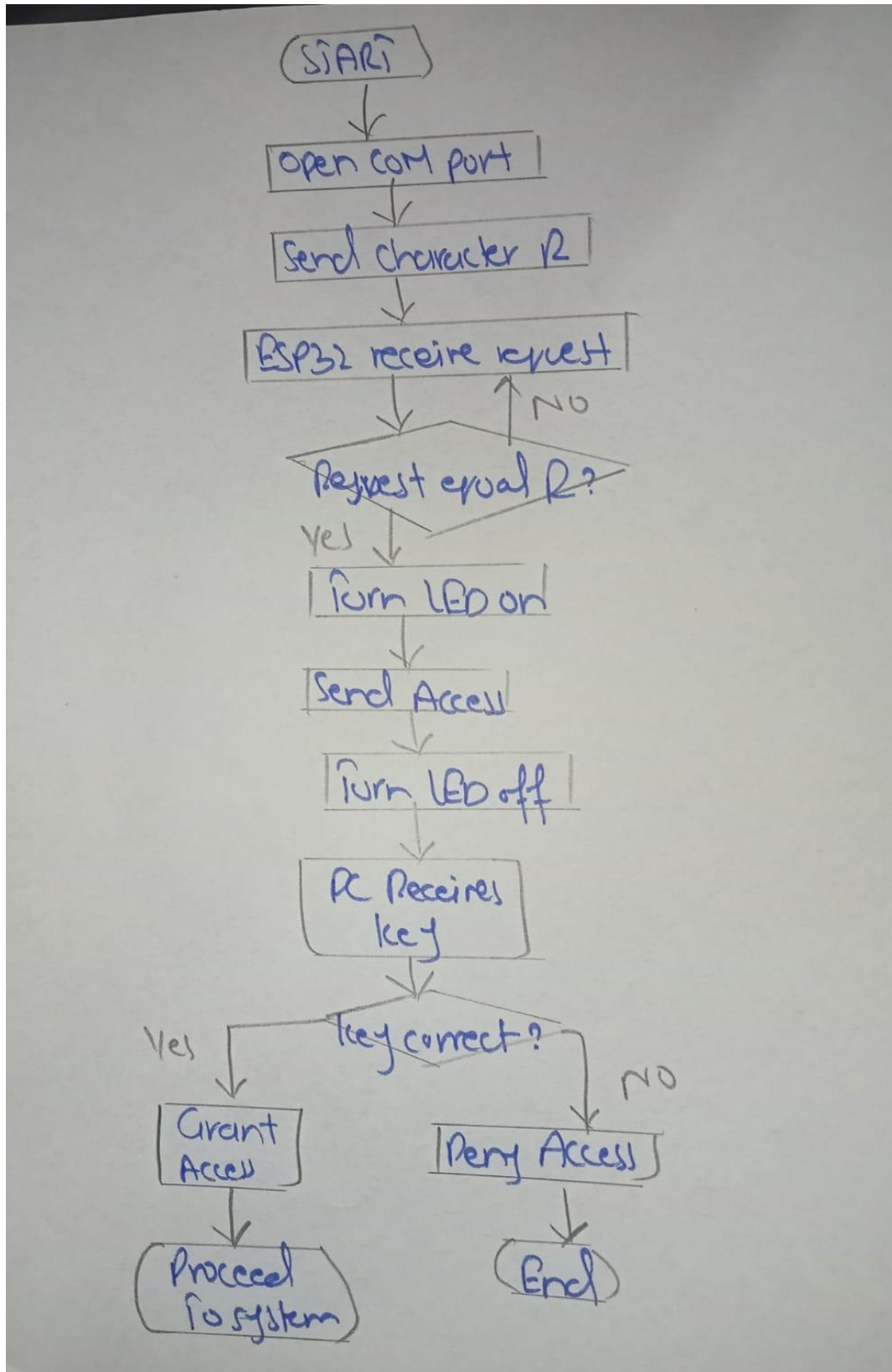
Flowchart For Add Student Record:



Flowchart For Search, Update and Delete Operations:



Flowchart For ESP32 Authentication Flow:



Deliverables with Complete Specification Sheet:

Deliverables:

- Student Registration Module
- Attendance Storage Module
- Search Module
- Update Module
- Delete Module
- Display Module
- CSV Database Storage

Specification Sheet:

Feature	Description
Programming Language	C++
Storage Method	CSV File
Interface	Console-Based
Duplicate Detection	Yes
Input Validation	Yes
Search Functionality	Yes
Update Functionality	Yes
Delete Functionality	Yes
Permanent Storage	Yes

Chapter 3 – Software Design

Input Output Details:

Inputs:

The system accepts the following user inputs

- Student ID
- Student Name
- Department Name
- Attendance Percentage

Outputs:

The system generates:

- Confirmation messages
- Student information records
- Attendance reports
- Validation error messages

Controller Selection with Features:

Since this project is software-based, the computer system acts as the controller.

Features

- Fast processing
- Reliable execution
- Easy maintenance
- Efficient storage management
- User-friendly operation

Software Design Details and User Requirements:

User Requirements:

The user should be able to:

- Add student records.
- Search student records.
- Update student information.
- Delete records.

- Display all records.

Function Descriptions:

initializeDatabase()

Creates attendance.csv if it does not already exist and inserts column headers.

isUnique()

Checks whether the entered student ID already exists in the database.

appendRecord()

Adds a new student record to the CSV file.

searchById()

Searches for a student record using the student ID.

updateRecord()

Updates an existing student record using a temporary file method.

deleteRecord()

Removes a student record from the database.

displayAllRecords()

Displays every record stored in attendance.csv.

User Cases (Test Cases):

Test Case 1: Add Student

Input:

```
===== Student Attendance Management System =====  
1. Add Student Record  
2. Search Student Record  
3. Update Student Record  
4. Delete Student Record  
5. Display All Records  
6. Exit  
Enter your choice: 1  
Enter Student ID: AU101  
Enter Student Name: SALMAN  
Enter Department: MECHANICAL  
Enter Attendance Percentage: 58
```

Output:

```
Record added successfully.
```

Test Case 2: Duplicate Student ID

Input:

```
===== Student Attendance Management System =====  
1. Add Student Record  
2. Search Student Record  
3. Update Student Record  
4. Delete Student Record  
5. Display All Records  
6. Exit  
Enter your choice: 1  
Enter Student ID: AU101
```

Output:

```
Duplicate ID not allowed.
```


Test Case 3: Search Student

Input:

```
===== Student Attendance Management System =====  
1. Add Student Record  
2. Search Student Record  
3. Update Student Record  
4. Delete Student Record  
5. Display All Records  
6. Exit  
Enter your choice: 2  
Enter Student ID to search: AU101
```

Output:

```
===== Student Record Found =====  
AU101, SALMAN, MECHANICAL, 58
```

Test Case 4: Invalid Attendance

Input:

```
===== Student Attendance Management System =====  
1. Add Student Record  
2. Search Student Record  
3. Update Student Record  
4. Delete Student Record  
5. Display All Records  
6. Exit  
Enter your choice: 1  
Enter Student ID: AU102  
Enter Student Name: ASLAM  
Enter Department: CHEMICAL  
Enter Attendance Percentage: 120
```

Output:

```
Invalid Attendance Percentage.
```

Test Case 5: Delete Student

Input:

```
===== Student Attendance Management System =====  
1. Add Student Record  
2. Search Student Record  
3. Update Student Record  
4. Delete Student Record  
5. Display All Records  
6. Exit  
Enter your choice: 4  
Enter Student ID to delete: AU101
```

Output:

```
Record deleted successfully.
```

Test Case 6: Display all Records

Input:

```
===== Student Attendance Management System =====  
1. Add Student Record  
2. Search Student Record  
3. Update Student Record  
4. Delete Student Record  
5. Display All Records  
6. Exit  
Enter your choice: 5
```

Output:

```
===== All Student Records =====  
  
ID, Name, Department, Attendance  
AU001, Zain Siddiqui, Electrical Engineering, 84  
AU002, Hassan Chaudhry, Cyber Security, 87  
AU003, Uzair Ali, Information Technology, 94  
AU004, Saad Iqbal, Software Engineering, 87  
AU005, Shayan Sheikh, Civil Engineering, 76  
AU006, Hassan Tariq, Cyber Security, 71  
AU007, Ayan Ali, Electrical Engineering, 92  
AU008, Ayan Akhtar, Artificial Intelligence, 83  
AU009, Saad Nawaz, Information Technology, 89  
AU010, Zain Qureshi, Data Science, 72  
AU011, Faizan Aslam, Electrical Engineering, 81  
AU012, Hassan Butt, Data Science, 96  
AU013, Shayan Akhtar, Information Technology, 100  
AU014, Faizan Nawaz, Business Administration, 78  
AU015, Ahmed Iqbal, Computer Science, 82  
AU016, Farhan Ali, Civil Engineering, 77  
AU017, Talha Khan, Software Engineering, 91  
AU018, Usman Nawaz, Mechanical Engineering, 84  
AU019, Usman Aslam, Computer Science, 90  
AU020, Hussain Akhtar, Business Administration, 91  
AU021, Ahmed Ali, Mechanical Engineering, 85  
AU022, Ayan Raza, Information Technology, 95  
AU023, Abdullah Siddiqui, Civil Engineering, 94  
AU024, Daniyal Raza, Cyber Security, 73  
AU025, Abdullah Chaudhry, Electrical Engineering, 73  
AU026, Ayan Raza, Cyber Security, 92  
AU027, Hussain Ahmed, Mechanical Engineering, 96  
AU028, Umer Akhtar, Electrical Engineering, 83  
AU029, Saad Nawaz, Computer Science, 85  
AU030, Ahmed Aslam, Business Administration, 70  
AU031, Talha Malik, Artificial Intelligence, 79  
AU032, Shayan Nawaz, Business Administration, 81
```

Deliverable of Complete Commented Code:

This system is made by using modular programming rules. Each operation is performed through separate function to improve the maintainability and readability of the system.

Programming Concepts Used:

- Functions
- Loops
- Conditional Statements
- File Handling
- String Manipulation
- CRUD Operations
- Input Validation

Chapter 4 – System Test Phase

Final Testing:

Function	Status
Database Initialization	Passed
Add Student	Passed
Search Student	Passed
Update Student	Passed
Delete Student	Passed
Display Records	Passed
Duplicate Detection	Passed
Attendance Validation	Passed

Problems Encountered:

Problem 1: Duplicate Records

Duplicate student IDs caused inconsistencies.

Solution: Implemented ID uniqueness checking.

Problem 2: Updating Records

Direct modification inside CSV files was difficult.

Solution: Temporary file replacement technique was used.

Problem 3: Invalid Attendance Values

Users could enter values beyond valid limits.

Solution: Attendance validation range (0–100%) was implemented.

Project Actual Pictures:

Main Menu

```
===== Student Attendance Management System =====
1. Add Student Record
2. Search Student Record
3. Update Student Record
4. Delete Student Record
5. Display All Records
6. Exit
Enter your choice:
```

Add Student Record Screen

```
===== Student Attendance Management System =====
1. Add Student Record
2. Search Student Record
3. Update Student Record
4. Delete Student Record
5. Display All Records
6. Exit
Enter your choice: 1
Enter Student ID: AU101
Enter Student Name: SAIFF
Enter Department: MECHATRONICS
Enter Attendance Percentage: 45
Record added successfully.
```

Attendance.csv File

```
Mainn.cpp  attendance.csv X
E: > Student_Attendance_Management_System > attendance.csv
94  AU093,Hussain Nawaz,Artificial Intelligence,75
95  AU094,Talha Khan,Artificial Intelligence,87
96  AU095,Talha Sheikh,Mechanical Engineering,77
97  AU096,Hassan Iqbal,Electrical Engineering,99
98  AU097,Farhan Aslam,Business Administration,84
99  AU098,Faizan Iqbal,Electrical Engineering,78
100 AU099,Hassan Chaudhry,Software Engineering,87
101 AU100,Usman Ahmed,Mechanical Engineering,100
102 AU101,SAIFF,MECHATRONICS,45
```

Search Student Screen

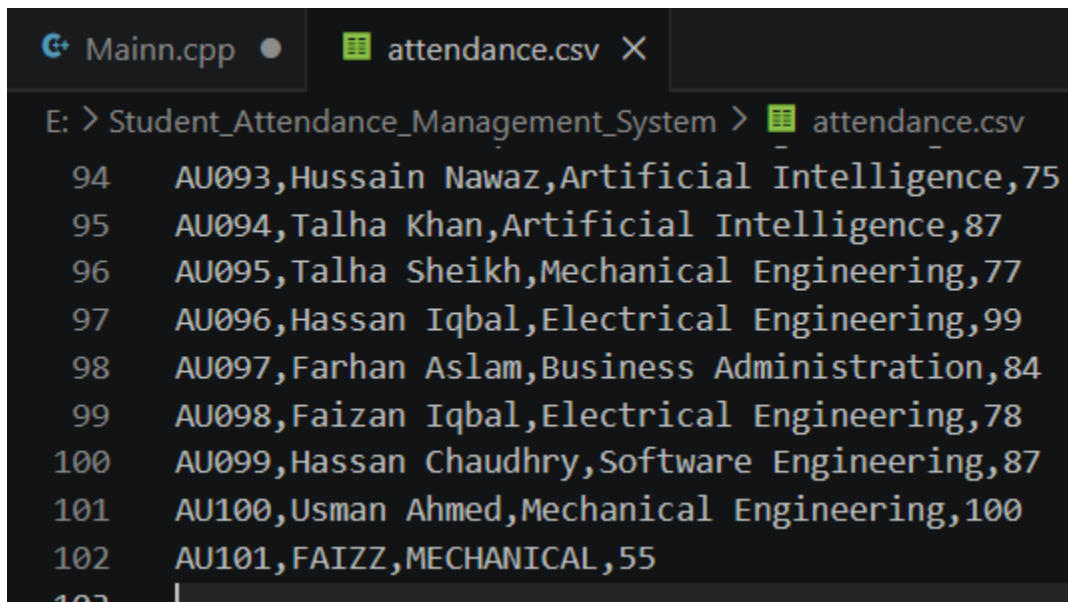
```
===== Student Attendance Management System =====
1. Add Student Record
2. Search Student Record
3. Update Student Record
4. Delete Student Record
5. Display All Records
6. Exit
Enter your choice: 2
Enter Student ID to search: AU101

===== Student Record Found =====
AU101,SAIFF,MECHATRONICS,45
```

Update Student Screen

```
===== Student Attendance Management System =====
1. Add Student Record
2. Search Student Record
3. Update Student Record
4. Delete Student Record
5. Display All Records
6. Exit
Enter your choice: 3
Enter Student ID to update: AU101
Enter New Name: FAIZZ
Enter New Department: MECHANICAL
Enter New Attendance Percentage: 55
Record updated successfully.
```

Attendance.csv File



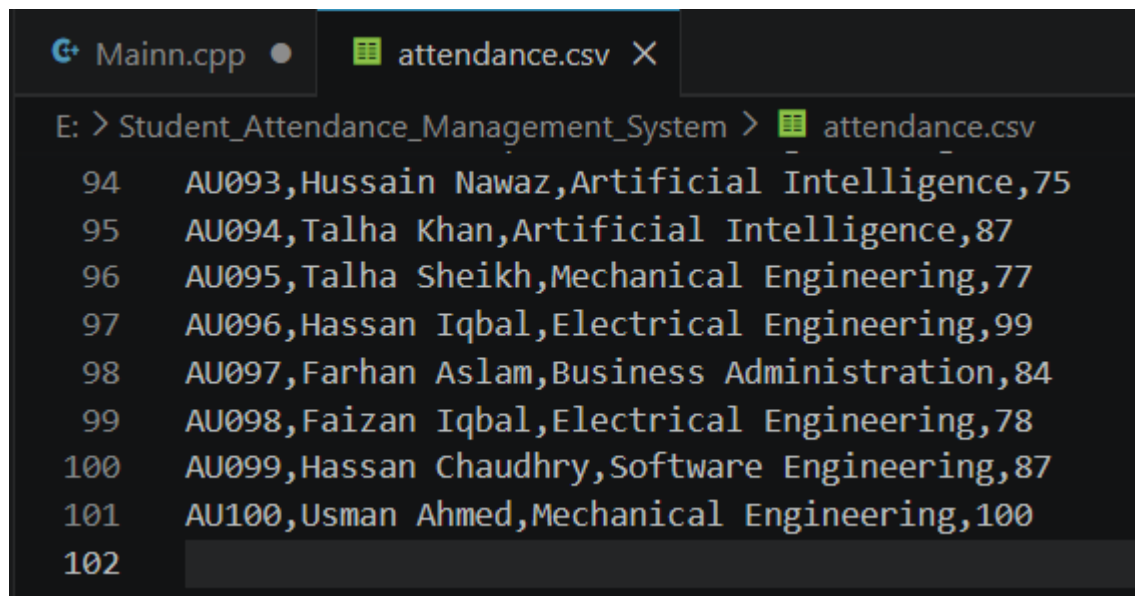
The screenshot shows a code editor with two tabs: 'Mainn.cpp' and 'attendance.csv'. The 'attendance.csv' tab is active, displaying the following content:

```
E: > Student_Attendance_Management_System > attendance.csv
94 AU093,Hussain Nawaz,Artificial Intelligence,75
95 AU094,Talha Khan,Artificial Intelligence,87
96 AU095,Talha Sheikh,Mechanical Engineering,77
97 AU096,Hassan Iqbal,Electrical Engineering,99
98 AU097,Farhan Aslam,Business Administration,84
99 AU098,Faizan Iqbal,Electrical Engineering,78
100 AU099,Hassan Chaudhry,Software Engineering,87
101 AU100,Usman Ahmed,Mechanical Engineering,100
102 AU101,FAIZZ,MECHANICAL,55
103 |
```


Delete Student Screen

```
===== Student Attendance Management System =====  
1. Add Student Record  
2. Search Student Record  
3. Update Student Record  
4. Delete Student Record  
5. Display All Records  
6. Exit  
Enter your choice: 4  
Enter Student ID to delete: AU101  
Record deleted successfully.
```

Attendance.csv File



The screenshot shows a code editor with two tabs: 'Mainn.cpp' and 'attendance.csv'. The 'attendance.csv' tab is active, displaying the following content:

```
E: > Student_Attendance_Management_System > attendance.csv  
94 AU093,Hussain Nawaz,Artificial Intelligence,75  
95 AU094,Talha Khan,Artificial Intelligence,87  
96 AU095,Talha Sheikh,Mechanical Engineering,77  
97 AU096,Hassan Iqbal,Electrical Engineering,99  
98 AU097,Farhan Aslam,Business Administration,84  
99 AU098,Faizan Iqbal,Electrical Engineering,78  
100 AU099,Hassan Chaudhry,Software Engineering,87  
101 AU100,Usman Ahmed,Mechanical Engineering,100  
102
```

Display Records Screen

```
===== Student Attendance Management System =====
```

1. Add Student Record
2. Search Student Record
3. Update Student Record
4. Delete Student Record
5. Display All Records
6. Exit

```
Enter your choice: 5
```

```
===== All Student Records =====
```

```
ID,Name,Department,Attendance
AU001,Zain Siddiqui,Electrical Engineering,84
AU002,Hassan Chaudhry,Cyber Security,87
AU003,Uzair Ali,Information Technology,94
AU004,Saad Iqbal,Software Engineering,87
AU005,Shayan Sheikh,Civil Engineering,76
AU006,Hassan Tariq,Cyber Security,71
AU007,Ayan Ali,Electrical Engineering,92
AU008,Ayan Akhtar,Artificial Intelligence,83
AU009,Saad Nawaz,Information Technology,89
AU010,Zain Qureshi>Data Science,72
AU011,Faizan Aslam,Electrical Engineering,81
AU012,Hassan Butt>Data Science,96
AU013,Shayan Akhtar,Information Technology,100
AU014,Faizan Nawaz,Business Administration,78
AU015,Ahmed Iqbal,Computer Science,82
AU016,Farhan Ali,Civil Engineering,77
AU017,Talha Khan,Software Engineering,91
AU018,Usman Nawaz,Mechanical Engineering,84
AU019,Usman Aslam,Computer Science,90
AU020,Hussain Akhtar,Business Administration,91
AU021,Ahmed Ali,Mechanical Engineering,85
AU022,Ayan Raza,Information Technology,95
AU023,Abdullah Siddiqui,Civil Engineering,94
AU024,Daniyal Raza,Cyber Security,73
AU025,Abdullah Chaudhry,Electrical Engineering,73
AU026,Ayan Raza,Cyber Security,92
AU027,Hussain Ahmed,Mechanical Engineering,96
```

attendance.csv Database File

```
Mainn.cpp ● attendance.csv X
E: > Student_Attendance_Management_System > attendance.csv
1 ID,Name,Department,Attendance
2 AU001,Zain Siddiqui,Electrical Engineering,84
3 AU002,Hassan Chaudhry,Cyber Security,87
4 AU003,Uzair Ali,Information Technology,94
5 AU004,Saad Iqbal,Software Engineering,87
6 AU005,Shayan Sheikh,Civil Engineering,76
7 AU006,Hassan Tariq,Cyber Security,71
8 AU007,Ayan Ali,Electrical Engineering,92
9 AU008,Ayan Akhtar,Artificial Intelligence,83
10 AU009,Saad Nawaz,Information Technology,89
11 AU010,Zain Qureshi,Data Science,72
12 AU011,Faizan Aslam,Electrical Engineering,81
13 AU012,Hassan Butt,Data Science,96
14 AU013,Shayan Akhtar,Information Technology,100
15 AU014,Faizan Nawaz,Business Administration,78
16 AU015,Ahmed Iqbal,Computer Science,82
17 AU016,Farhan Ali,Civil Engineering,77
18 AU017,Talha Khan,Software Engineering,91
19 AU018,Usman Nawaz,Mechanical Engineering,84
20 AU019,Usman Aslam,Computer Science,90
21 AU020,Hussain Akhtar,Business Administration,91
22 AU021,Ahmed Ali,Mechanical Engineering,85
23 AU022,Ayan Raza,Information Technology,95
24 AU023,Abdullah Siddiqui,Civil Engineering,94
25 AU024,Daniyal Raza,Cyber Security,73
26 AU025,Abdullah Chaudhry,Electrical Engineering,73
27 AU026,Ayan Raza,Cyber Security,92
28 AU027,Hussain Ahmed,Mechanical Engineering,96
29 AU028,Umer Akhtar,Electrical Engineering,83
30 AU029,Saad Nawaz,Computer Science,85
31 AU030,Ahmed Aslam,Business Administration,70
32 AU031,Talha Malik,Artificial Intelligence,79
33 AU032,Shayan Nawaz,Business Administration,81
34 AU033,Shayan Ahmed,Computer Science,87
35 AU034,Daniyal Butt,Information Technology,86
36 AU035,Hassan Ahmed,Electrical Engineering,72
37 AU036,Umer Khan,Cyber Security,79
```

Chapter 5 – Project Management

How Each Member Executed His Role in this Project:

Soban Akhtar – Team Lead

As the team leader of this project, I was responsible for supervising all the project process and ensures that all project objectives were complete successfully. I coordinated meetings with other team members to discuss the requirements, objectives and how to implement the strategies. I also distribute the tasks to other members according to their decided roles and monitored them everyday to check that the project is going faster or not. I also participate in requirement analysis and help to define the scope of project. I also solved the issues that arises across the development by discussing with my team members. I also revised the work of my members to check that all the activities were fully completed with the specific timeline. I also supervised the documentation preparation and verified thar the final reports are made according to the requirements given. I also contributed to testing and evaluation the activities to make sure that the system works according to the requirements. Through proper planning and coordination with my members I helped to maintain the project quality and make sure that the project completed successfully without any errors

Abdullah Khan – Project Manager

As the project Manager my task is to plan, organize and manage project activities from start to end. My primary responsibility was to prepare a schedule and make sure that project task was completed according to the deadlines mentioned in the schedule. I also talked with team members regularly to monitor the progress and address any challenges face during the making of the system. I also take place in project planning, requirement gathering and how to design the system. I maintain records of completed tasks and make sure that project milestones were achieved completely. I also helped prepare documentation, presentations and progress reports. Another imp responsibility was to manage the resources of project and communication among team members. My role helped that all the planned objectives were completed in time and according to the required requirements

Uzair Iqbal – Technical Integrator & Tester

I was responsible for the integrating the software modules and verify that the complete management system works perfectly. After individual's module were made, I combine all of them into a single working system and ensure compatibility among all the components. I also conducted testing on all the functionalities that include adding, searching, updating, deleting, and displaying the student records. I make too many test cases to verify system reliability and identify the errors that occur during the process. Special attention was given to attendance validation and duplicate id detection. Whenever any bugs arise, I work closely with the team to solve them quickly and efficiently. I also verify that the Csv database file correctly stored the records and give attendance information correctly. By making systematic testing and procedures I ensured that the software met all the requirements that are given and delivered accurate results. Mu role contributed significantly to improve the system stability, reliability and remove any bugs and errors and to make sure that the system works perfectly without any problems

Individual Comments on Success / Failure of the Project:

Soban Akhtar – Team Lead

The project was successful because we planned that all the functionalities were executed and tested correctly. This management system performs different operation accurately and stores the records properly using the CSV file handling process. In future projects more time is needed for testing and making the document activities before the final submission of the project.

Abdullah Khan – Project Manager

It was the successful project achieving all the objectives that are planned in the schedule. The teamwork and task management were taking place properly and were effective throughout the whole project. Some minor challenges occurred during testing and making the project, but they were solved by the teamwork and by proper planning. Future projects benefit by progress reviews and by more detailed plan during early stages

Uzair Iqbal – Technical Integrator & Tester

By testing the results of the project conform that all the important features work accurately and fulfill the project requirements. The system properly avoids duplicate records and checks attendance percentages correctly. One drawback of the project is the absence of a visual user interface and connection with a complete database system. In Future updates these issues can solve and make the system easier and better for users to use.

Evaluation of Other Team Members:

Abdullah Khan's Evaluation

My team member Soban Akhtar works professionally and maintains the decorum of the schedule. He manages the schedule of the project and make an environment for smooth communication among other members. Another team member Uzair has strong abilities during the testing and implementing the activities. Both members were very supportive and cool and has the courage to achieve project objectives.

Soban Akhtar's Evaluation

My team member Abdullah maintains the good coordination among team members throughout the whole project. His planning and supervision make success to the project. Another team member Uzair perform the activities like testing and finding the errors or issues before final deployment. Both members works skilfully and both members are so supportive and cooperative

Uzair Iqbal Evaluation

My team partner Abdullah khan is great in managing the project smoothly and make sure that all the activities done according to the given requirements. Another partner Soban is also excellent at management and have skills to maintain the project and make sure that all the members work according to the given timeline. Both members are very good supportive and cooperative and helpful in every problem that arises during the work

Risk Management Learned:

The Student Attendance Management System project gives us the valuable experience for risk management and planning of the project. One of the most important lessons we learned was the importance of finding the risks during the early stages of development of the code. Risks include as data loss, coding errors, integration problems, schedule delays, and by not testing step by step can affect the project success if not handled properly. The project also explains us the importance of maintaining regular backups of project files and source code to prevent accidental data loss. By step-by-step testing of the development helped us to identify errors early and reduced the major issues during final deployment. The communication among team members minimized misunderstandings and improved coordination. Another important lesson we learned was the value of proper task management and time management. By giving the tasks to the team members according to his skills makes productivity and reduced project risks. At the end, planning, regular progress checkups, and thorough step by step testing proved essential for ensuring project quality and successful completion.

Chapter 6 – Feedback For Project and Course

The project gives practical experience in the software development, the file handling, testing the system and solving the problems of the management system. By making the student attendance management system we enhanced our understanding the C++ programming tips and tricks and the concepts including function, how to handle a file, conditional statements, loops and string manipulation also by doing this project our teamwork is improved our communication is improved, problem solving is improved and documentation skill also improved. This course successfully connects the theoretical knowledge with practical implementation and gives valuable life experience in managing a real software development project.

References:

- [1] K. Patel and M. Shah, "*Student Attendance Management System Using Digital Technology*," International Journal of Computer Applications, vol. 180, no. 25, pp. 15–20, 2018.
- [2] A. Kumar and R. Singh, "*Automated Attendance Management System Using Database Technology*," International Journal of Engineering Research & Technology (IJERT), vol. 9, no. 4, pp. 120–125, 2020.
- [3] S. Sharma and P. Verma, "*Design and Implementation of Student Information and Attendance Management System*," International Journal of Advanced Research in Computer Science, vol. 8, no. 5, pp. 432–437, 2017.
- [4] I. Sommerville, *Software Engineering*, 10th ed., Pearson Education, 2016.
- [5] B. Stroustrup, *Programming: Principles and Practice Using C++*, 2nd ed., Addison-Wesley, 2014.
- [6] H. M. Deitel and P. J. Deitel, *C++ How to Program*, 10th ed., Pearson Education, 2017.
- [7] M. Fowler, *UML Distilled: A Brief Guide to the Standard Object Modeling Language*, 3rd ed., Addison-Wesley, 2003.
- [8] R. Pressman and B. Maxim, *Software Engineering: A Practitioner's Approach*, 8th ed., McGraw-Hill Education, 2015.
- [9] Microsoft Documentation, *File Handling and Data Storage Techniques in C++*, Microsoft Learn
- [10] ISO/IEC 14882:2020, *Programming Languages — C++*, International Organization for Standardization (ISO), Geneva, Switzerland, 2020